

Evals Interview Tips

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Takehome Exam: You Get A Dataset. “Do Evals”

**Claude - find all errors.
Make no mistakes.**



Our example: Tree Buddy



Guardrails

Tree facts

Health findings

Summary

Recommendation

It's really a data challenge

- Labeled Data:
 - Guardrails (pass/fail): safety, out-of-bounds inputs or outputs
 - Data Extraction: species of tree, age, health, etc.
- Unlabeled Data
 - Text: summary, recommendation
- Build an agent that can **perform “well”** w/evals!

Here's my structure

1. Executive Summary
2. Data
3. Experiments
4. Results
5. What I Learned
6. Next Steps
7. AI Tools Used

Executive Summary

Keep it to 1/3rd of a page or less. What you built and the results.

Short discussion on technical wins and limitations.

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Data

Spend 50-70% of your time here. Take note of issues

- Missing or inconsistent labels.
- Insufficient labeled data
- How noisy is the data?

Data - Missing Labels

Target	Labeled / 120	Violations
Species	120	—
Health finding	96	—
Height	24	—
Age	8	—
Input policy	120	0

Can outside data fill the missing labels?

Research Data Archive
Roots of our Research



Search catalog

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Publication Details

Title: Urban tree database
Author(s): [McPherson, E. Gregory](#); [van Doorn, Natalie S.](#); [Peper, Paula J.](#)
Publication Year: 2016
How to Cite: These data were collected using funding from the U.S. Government and can be used without additional permissions or fees. If you use these data in a publication, presentation, or other research product please use the following citation:

McPherson, E. Gregory; van Doorn, Natalie S.; Peper, Paula J. 2016. Urban tree database. Fort Collins, CO: Forest Service Research Data Archive. Updated 21 January 2020. <https://doi.org/10.2737/RDS-2016-0005>

Abstract: This data publication contains urban tree growth data collected over a period of 14 years (1998-2012) in 17 cities from 13 states across the United States: Arizona, California, Colorado, Florida, Hawaii, Idaho, Indiana, Minnesota, New Mexico, New York, North Carolina, Oregon, and South Carolina.

Measurements were taken on over 14,000 urban street and park trees. Key information collected for each tree species includes bole and crown size, location, and age. Based on these measurements, 365 sets of allometric equations were developed for tree species from around the U.S. Each "set" consists of eight equations for each of the approximately 20 most abundant species in each of 16 climate regions. Tree age is used to predict a species diameter at breast height (dbh), and dbh is used to predict tree height, crown diameter, crown height, and leaf area. Dbh is also used to predict age. For applications with remote sensing, average crown diameter is used to predict dbh. There are 171 distinct species represented within this database. Some species grow in more than one region. The Urban Tree Database (UTD) contains foliar biomass data (raw data and summarized results from the foliar sampling for each species and region) that are fundamental to calculating leaf area, as well as tree biomass equations (compiled from literature) for carbon storage estimates. An expanded list of dry weight biomass density factors for common urban species is made available to assist users in using volumetric equations.

 **Hugging Face**

[Models](#) [Datasets](#) [Spaces](#) [Bugs](#)

Datasets: [BreadboardLabs/CurioTreeData](#)  Like 0 [Follow](#)  Curio 3

Size: 1M< n<10M Tags: [climate](#) [trees](#) [images](#) License:  cc-by-nc-4.0

[Dataset card](#) [Data Studio](#) [Files and versions](#) [xet](#) [Community](#)

Dataset Preview  API  Embed  Duplicate  Data Studio

Split (1)
train

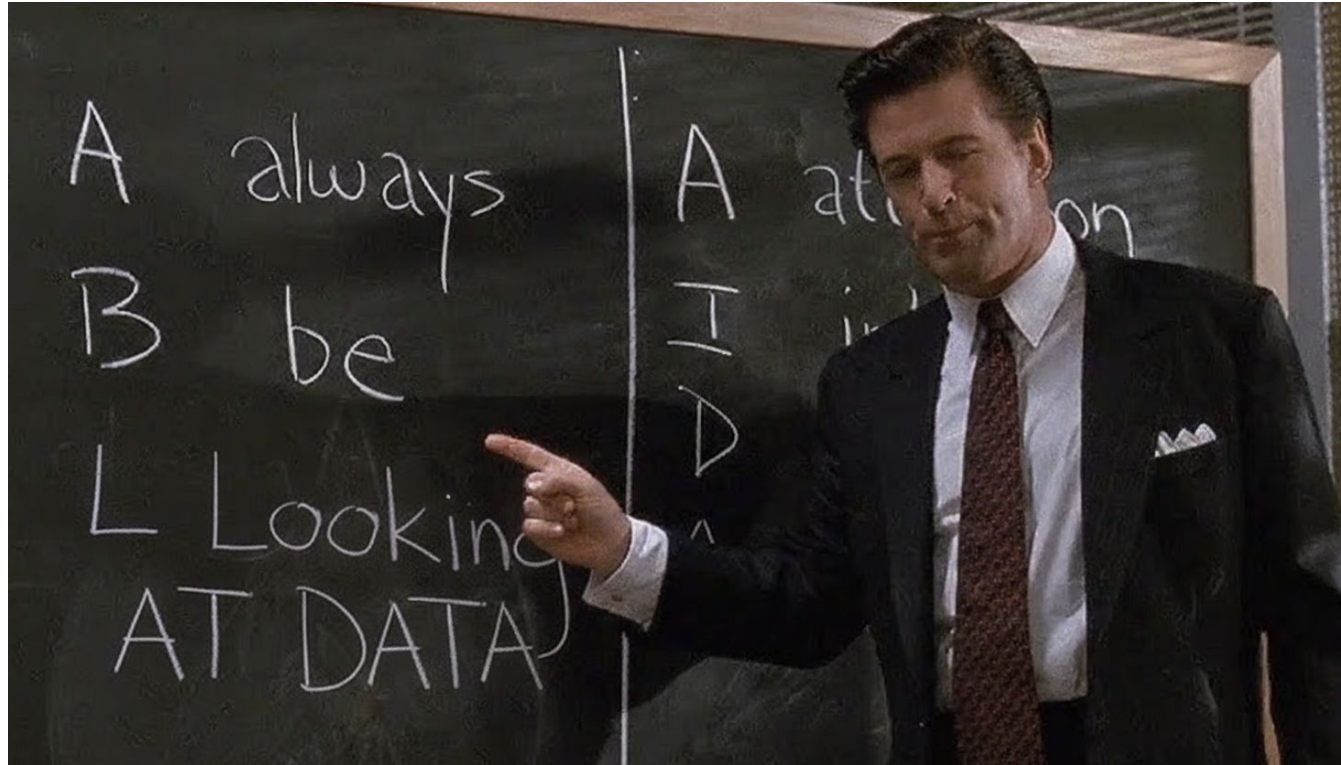
The full dataset viewer is not available (click to read why). Only showing a preview of the rows.

id	user_id	conversation_id	content	status	created_at	updated_at
int64	int64	int64	string	string	string	string
2	179	2	The owner was out watering her plants and kindly told me tha...	PUBLISHED	2017-05-24 16:16:58 UTC	2017-05-24 16:17:14 UTC
24	284	16	Yeah, definitely, they look a lot like big bonsai trees!	PUBLISHED	2017-05-29 18:56:15 UTC	2017-05-29 19:16:49 UTC
3	178	3	🌳🌳🌳	PUBLISHED	2017-05-24 16:24:19 UTC	2017-05-24 16:33:04 UTC
4	180	4	This looks beautiful 🌳. I'm so used to seeing hawthorn...	PUBLISHED	2017-05-24 16:38:46 UTC	2017-05-24 17:15:29 UTC
1	178	1	That's a lovely example of a hawthorn.	PUBLISHED	2017-05-24 15:45:36 UTC	2017-05-24 17:15:55 UTC
5	179	5	Is that a telephone cable running right through the...	PUBLISHED	2017-05-24 17:30:39 UTC	2017-05-24 17:31:48 UTC
6	180	5	It surely is #overheadcable	PUBLISHED	2017-05-24 17:32:16 UTC	2017-05-24 17:32:16 UTC

Guardrails: Add examples of what to block and allow



Look at the actual records



Data - Inconsistent Labels



Species: English oak
Height: 12 m



Species: "Pine.Long needles, orange bark."
Height: "50"

Data - Noise



Summarize The Data Issues

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Experiment: Three parts

1. Goal
2. What you tried
3. Methods & Findings

Goal: species identification

Classify the species correctly.

What I tried: species identification

Run	Model	Notes	Val. F1	Test F1
V1	GPT-6 Luna	Prompt only	0.657	0.626
V2	GPT-6 Luna	RAG: external tree database	0.788	0.760
V3	GPT-6 Sol	Prompt only	0.817	0.788
V4	GPT-6 Sol	RAG: external tree database	0.927	0.899
V5	GPT-6 Astra	Prompt only	0.948	0.682
V6	GPT-6 Astra	RAG: external tree database	0.971	0.677
V7	GPT-6 Sol	RAG + automated prompt tuning	0.899	0.879

Findings & Methods

Methods:

- explain retrieval, prompt tuning.

Findings:

- Sol + RAG seems like the best tradeoff.
- Astra's is overfitting.
- Automated prompt tuning did not help.



Experiment: If you are asked to do the wrong thing



“The brown leaf edges mean your tree is drought-stressed. Give it extra water this week and watch for more browning. It’s clear you have been neglecting your tree!”

**Grade helpfulness and tone from 1–5,
combine them into a composite score of
0-100.**

Recommend using specific pass/fail checks

- Is full length picture taken?
- Watering advice checks soil moisture
- No blame or scolding

Describe WHY these are better

You: "We check our AI with coherence score"

Them: "How we can trust this coherence score?"

You:



Experiments: Put It All Together

Goal

Experiment

Methods + Findings

There will be lots of experiments!

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Results

Synthesize your experiments

1. Your recommendation across all experiments
2. Baseline vs. Final Eval Scores summarized
3. Cost, Latency tradeoffs

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What I learned

1. What worked well

- a. Synthetic Data, External Data + Retrieval, Train/Val/Test caught overfitting

2. What didn't work

- a. Prompt tuning, larger models

3. Limitations

- a. Repeat any “bad” things you were forced to do very diplomatically. “LLM Judges were not validated against human labels because of XYZ”
- b. Repeat any limitations of the data or the approach



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Next Steps

- Ideas for improving performance
- Always: Better data, product improvements
 - Discuss details

Questions?